Peng Cheng

pengcheng326@hotmail.com • (86) 15057169279 • https://pengcheng-tech.github.io/ • Hangzhou, Zhejiang, China

Research Interests: AIGC security, IoT security

Professional Experience

Researcher, State Key Laboratory of Blockchain and Data Security, Zhejiang University 2024-06 — Present
Member of the AI Data Security Team

- Conducting research on AI-generated content (AIGC) security
- Developing the DFscan platform for multimodal deepfake detection; leading the technical team in platform design and development

Qiushi Research Fellow, ZJU-Hangzhou Global Scientific and Technological Innovation Center 2023-12 — 2024-05
Member of the Cyberspace Security Research Institute

- Conducted research on multimodal data security and privacy protection technologies
- Developed multimodal deepfake detection systems

Postdoctoral Researcher, Zhejiang University 2021-01 — 2023-11
College of Computer Science and Technology

- Researched voice security and privacy protection in human-computer interaction scenarios
- Postdoctoral supervisor: Prof. Kui Ren (Qiushi Chair Professor, AAAS/ACM/CCF/IEEE Fellow, Dean of the College of Computer Science and Technology of Zhejiang University)

Education

Ph.D., Computer Science — Lancaster University, Lancaster, UK
2016-10 — 2020-12
School of Computing and Communications. Thesis: Acoustic-Channel Attack and Defence Methods for Personal Voice Assistants. Supervisors: Prof. Utz Roedig (University College Cork) and Prof. Jeff Yan (University of Southampton)

Dual Master's Degree, Electrical Engineering (KU Leuven) & Integrated Circuit Engineering (Tsinghua) — KU Leuven & Tsinghua University, Leuven, Belgium & Beijing, China
2012-09 — 2015-09
Department of Electrical Engineering, KU Leuven & School of Integrated Circuits, Tsinghua University. Supervisor: Prof. Guoqiang Bai (Tsinghua University)

B.Eng., Electronic Science and Technology — Beijing University of Posts and Telecommunications, Beijing, China
2008-09 — 2012-06
School of Electronic Engineering

Research Projects

National Natural Science Foundation of China (NSFC)

- Research on Speech Synthesis Data Compliance Management Technology Based on Intrinsic Characteristics of Audio Signals, (Principal Investigator), 2025 — 2028
- High-Performance Visual Perception Models Using Deep Learning, (Participant), 2023 — 2026
- Cross-Chain Security in Heterogeneous Blockchain Networks, (Participant), 2023 — 2027
- Research on Voice Attack and Defense Based on the Physical Characteristics of Smart Device Sensing Components, (Participant), 2022 — 2025

Key R&D Programs of China

- Multimodal Network Environment Construction Technology Based on Public Cloud-Network Resources, (Participant), 2024 — 2027
- Aggregation and Transfer of Machine Learning Models (National Science and Technology Innovation 2030 Initiative - New Generation Artificial Intelligence), (Participant), 2021 — 2025
- Security Protection Technology for Industrial Control Programming Platforms Based on Domestic Cryptographic Algorithms, (Participant), 2022 — 2024
- Multidimensional Protocol Carrying and Endogenous Security Access Technologies for Multimodal Networks, (Participant), 2024 — 2027

Provincial, Municipal, and University-Level Projects

- Key R&D Programme of Zhejiang Province, (Participant), 2024 — 2026

Hangzhou Key R&D Program

- Key Technologies and Platform Development for Security Detection of Large AI Models, (Participant), 2024 — 2027

Hangzhou West Innovation Corridor Development Special Fund

- Toolchain for Deep Synthetic Content Analysis Based on Physical Attribute Attribution, (Principal Investigator), 2024 — 2026

Zhejiang University-Alibaba Collaboration Project

- Active and Passive Security Protection Technologies for the Maojing Voice Interaction System, (Principal Investigator), 2025 — 2026

Zhejiang University-Ant Group Joint Laboratory

- Security Risk Detection and Alignment Strategies for Large Model-based AI Agents, (Participant), 2025 — 2025

China Southern Power Grid Research Institute Co., Ltd.

- Research and Development of AI-Driven Automated Security Detection Technologies and Components for Power Systems, (Participant), 2024 — 2026
- Research on AI Attack-Defense Library Design and Test Component Development for New Power System Scenarios (2023), (Participant), 2023 — 2025

CRRC Zhuzhou Electric Locomotive Research Institute Co., Ltd.

- Deep Learning Algorithm Evaluation and Security Verification Platform, (Participant), 2024 — 2027

Publications

Representative Publications

- Zhongjie Ba, Jieming Zhong, Jiachen Lei, **Peng Cheng**, Qinglong Wang, Zhan Qin, Zhibo Wang, Kui Ren (2024). SurrogatePrompt: Bypassing the Safety Filter of Text-to-Image Models via Substitution. Proceedings of the ACM SIGSAC Conference on Computer and Communications Security (CCS 2024), 1166—1180.
- **Peng Cheng**, Yuwei Wang, Peng Huang, Zhongjie Ba, Xiaodong Lin, Feng Lin, Li Lu, Kui Ren (2024). ALIF: Low-Cost Adversarial Audio Attacks on Black-Box Speech Platforms Using Linguistic Features. 2024 IEEE Symposium on Security and Privacy (SP), 1628—1645.
- Zhongjie Ba, Qing Wen, **Peng Cheng**, Yuwei Wang, Feng Lin, Li Lu, Zhenguang Liu (2023). Transferring Audio Deepfake Detection Capability Across Languages. Proceedings of the ACM Web Conference 2023, 2033—2044.
- **Peng Cheng**, Yuexin Wu, Yuan Hong, Zhongjie Ba, Feng Lin, Li Lu, Kui Ren (2024). UniAP: Protecting Speech Privacy With Non-Targeted Universal Adversarial Perturbations. IEEE Transactions on Dependable and Secure Computing, 31—46.
- **Peng Cheng**, Utz Roedig (2022). Personal Voice Assistant Security and Privacy—A Survey. Proceedings of the IEEE, 476—507.

Other Publications (Reverse Chronological Order)

- Qing Wen, Haohao Li, Zhongjie Ba, **Peng Cheng**, Miao He, Li Lu, Kui Ren (2026). HyperPotter: Spell the Charm of High-Order Interactions in Audio Deepfake Detection. International Conference on Machine

Learning (ICML 2026).

- Qingcao Li, Yipeng Lin, Weichen Lian, Zhongjie Ba, **Peng Cheng**, Zhichao Lian (2026). MixFake: Benchmarking and Enhancing Audio Deepfake Detection in Diverse Real-world Mixed Audio. IEEE International Conference on Multimedia and Expo (ICME 2026).
- Qingyu Liu, Yitao Zhang, Zhongjie Ba, Chao Shuai, **Peng Cheng**, Tianhang Zheng, Zhibo Wang (2026). Attack-Resistant Watermarking for AIGC Image Forensics via Diffusion-based Semantic Deflection. The 14th International Conference on Learning Representations (ICLR 2026).
- Zhongjie Ba, Liang Yi, **Peng Cheng**, Qingcao Li, Qinglong Wang, Li Lu (2026). Beyond Content: A Comprehensive Speech Toxicity Dataset and Detection Framework Incorporating Paralinguistic Cues. Proceedings of the AAAI Conference on Artificial Intelligence, 21—29.
- Kui Ren, Feng Lin, Zhongjie Ba, Zhuotao Liu, **Peng Cheng** (2025). Deepfake Detection: Key Challenges and Technical Approaches. Computing Magazine of the CCF, 8—15.
- Ziping Dong, Chao Shuai, Zhongjie Ba, **Peng Cheng**, Zhan Qin, Qinglong Wang, Kui Ren (2025). WMCopier: Forging Invisible Image Watermarks on Arbitrary Images. The 39th Conference on Neural Information Processing Systems (NeurIPS 2025).
- Peng Huang, Kun Pan, Qinglong Wang, **Peng Cheng**, Li Lu, Zhongjie Ba, Kui Ren (2025). SecHeadset: A Practical Privacy Protection System for Real-time Voice Communication. Proceedings of the 23rd Annual International Conference on Mobile Systems, Applications and Services (MobiSys 2025), 515—527.
- Peng Huang, Yao Wei, **Peng Cheng**, Zhongjie Ba, Li Lu, Feng Lin, Yang Wang, Kui Ren (2025). Phoneme-Based Proactive Anti-Eavesdropping With Controlled Recording Privilege. IEEE Transactions on Dependable and Secure Computing, 1924—1940.
- Zhe Xu, **Peng Cheng**, Zhongjie Ba, Peng Huang, Kui Ren (2025). Spoofed Speech Detection in Real-World Fraudulent Communication Scenarios. Journal of Cyber Security, 178—196.
- Zhongjie Ba, Bin Gong, Yuwei Wang, Yuxuan Liu, **Peng Cheng**, Feng Lin, Li Lu, Kui Ren (2024). Indelible "Footprints" of Inaudible Command Injection. IEEE Transactions on Information Forensics and Security, 8485—8499.
- Peng Huang, Yao Wei, **Peng Cheng**, Zhongjie Ba, Li Lu, Feng Lin, Fan Zhang, Kui Ren (2023). InfoMasker: Preventing Eavesdropping Using Phoneme-Based Noise. Proceedings 2023 Network and Distributed System Security Symposium.
- **Peng Cheng**, M. S. Arun Sankar, Ibrahim Ethem Bagci, Utz Roedig (2022). Adversarial Command Detection Using Parallel Speech Recognition Systems. Lecture Notes in Computer Science, 238—255.
- **Peng Cheng**, Ibrahim Ethem Bagci, Utz Roedig, Jeff Yan (2020). SonarSnoop: Active Acoustic Side-Channel Attacks. International Journal of Information Security, 213—228.
- **Peng Cheng**, Ibrahim Ethem Bagci, Jeff Yan, Utz Roedig (2019). Smart Speaker Privacy Control—Acoustic Tagging for Personal Voice Assistants. 2019 IEEE Security and Privacy Workshops (SPW), 144—149.
- **Peng Cheng**, Ibrahim Ethem Bagci, Jeff Yan, Utz Roedig (2018). Towards Reactive Acoustic Jamming for Personal Voice Assistants. Proceedings of the 2nd International Workshop on Multimedia Privacy and Security, 12—17.

Preprints / Under Review

- Qingcao Li, Miao He, Liang Yi, Qing Wen, Yitao Zhang, Hongshuo Jin, **Peng Cheng**, Zhongjie Ba, Li Lu, Kui Ren (2026). Divide and Conquer: Multimodal Video Deepfake Detection via Cross-Modal Fusion and Localization. arXiv preprint arXiv:2602.00209.
- Bin Gong, **Peng Cheng**, Zhongjie Ba, Li Lu, Jie Tang, Jinpeng Hou, Song Xue, Kui Ren (2025). Bridging the Synthesis-Detection Gap: A Chinese Audio Deepfake Dataset and Industrial-Scale Retrieval-Augmented Detection. Under review.
- Zhongjie Ba, Yitao Zhang, **Peng Cheng**, Bin Gong, Xinyu Zhang, Qinglong Wang, Kui Ren (2025). Robust Watermarks Leak: Channel-Aware Feature Extraction Enables Adversarial Watermark Manipulation. arXiv preprint arXiv:2502.06418.
- Bin Gong, Chao Shuai, Qing Wen, **Peng Cheng**, Qinglong Wang, Zhongjie Ba, Zhibo Wang, Kui Ren (2025). Test-Time Adaptation for Audio Deepfake Detection. Under review.
- Zhongjie Ba, Hao Fu, Yiming Yang, Hengzhi Chen, Qinglong Wang, **Peng Cheng**, Zhan Qin, Kui Ren (2025). JudgeRail: Harnessing Open-Source LLMs for Fast Harmful Text Detection with Judicial Prompting and Logit Rectification. Under review.

- Qing Wen, **Peng Cheng**, Zhongjie Ba, Liang Yi, Zhan Qin, Li Lu, Qinglong Wang, Kui Ren (2024). CLINDA: A Cross-lingual Domain Adaptation Framework for Challenging Audio Deepfake Detection Tasks across Languages. Under review.
- Jiachen Lei, Qinglong Wang, **Peng Cheng**, Zhongjie Ba, Zhan Qin, Zhibo Wang, Zhenguang Liu, Kui Ren (2023). Masked Diffusion Models Are Fast Distribution Learners. arXiv preprint arXiv:2306.11363.

Thesis

- **Peng Cheng** (2020). Acoustic-Channel Attack and Defence Methods for Personal Voice Assistants. Ph.D. Dissertation, Lancaster University.

Student Supervision and Mentorship

- Co-supervising 5 Ph.D. students, 7 Master's students and one undergraduate student at Zhejiang University on research projects in AIGC security and multimodal privacy
- Co-supervised a Zhejiang University Master's student to win the National Graduate Scholarship (China) — the highest-level scholarship awarded to Master's students in China (Oct 2024)
- Successfully mentored 1 PhD student, 5 Master's students, and 5 undergraduates to degree completion, guiding thesis design, research execution, and publication strategies
- Teaching Assistant for lab/practical sessions: CS4615: Computer Systems Security, University College Cork, Ireland; SCC110: Software Development, Lancaster University, UK

Honors \& Awards

- **Silver Award** | IJCAI 2026 Deepfake Detection and Localization Challenge (DDL 2.0) | (2026-08) | Team AIGVDete (Zhejiang University)
- **3rd Place Award** | Workshop & Challenge on Deepfake Detection, Localization, and Interpretability, IJCAI 2025 | (2025) | Advisor | Track: Audio-Video Detection and Localization (DDL-AV)
- **National Grand Prize (Top-Tier Award)** | 19th "Challenge Cup" National Competition for Extracurricular Academic Science and Technology Works | (2024) | Advisor (Ranked 2nd among 3 advisors) | Project: "Multimodal AI Audit Matrix: Deepfake Detection and NSFW Content Regulation Platform"
- **Top 5 Nationwide** | 3rd China Artificial Intelligence Competition | (2021) | Primary Advisor (Ranked 1st among 2 advisors) | Track: Audio Deepfake Detection Under Open-Speaker Scenarios
- **Top 5 Nationwide** | 3rd China Artificial Intelligence Competition | (2021) | Primary Advisor (Ranked 1st among 2 advisors) | Track: Speaker-Specific Audio Deepfake Detection
- **Finalist for "Most Innovative Research" Pwnie Award** | Black Hat USA | (2019) | First Author (1st of 4 contributors) | Research: "SonarSnoop: Active Acoustic Side-Channel Attacks"
- **Postdoctoral Excellence Grant (Second Class)** | Zhejiang Provincial Department of Human Resources and Social Security | (2021-08)
- **Ph.D. Scholarship** | Faculty of Science and Technology, Lancaster University, UK | (2016)

Academic Services

Journal Editorial Roles

- Special Issue Initiator and Guest Editor-in-Chief: "Intelligent Voice Security and Defense Technologies", Journal of Cyber Security, 2025

Conference Program Committees

- Program Committee Member: AAAI 2027, USENIX Security 2027

Conference Reviewer

- ICML 2026, ICLR 2026, ACM MM 2026, IEEE SLT 2026, Interspeech 2026, AAAI 2026, ACM Web Conference (WWW 2025)

Journal Reviewer Roles

- English Journals: Proceedings of the IEEE, IEEE Transactions on Information Forensics and Security (TIFS), IEEE Transactions on Dependable and Secure Computing (TDSC), IEEE Internet of Things Journal (IoT-J), ACM Transactions on Internet of Things (TIOT)
- Chinese Journal: Journal of Information Network Security

Reviewer Recognition

- ICML 2026 Silver Reviewer — top-tier recognition awarded by the ICML 2026 Program Chairs

Granted Patents

- Privacy Protection Method and Device Using White-Box Speech Adversarial Examples, Patent No. 2022109965917, CN 115208507 B, Granted: 2024-12-03, Rank 1/6
- Method and Device for Detecting Nonlinear Injection Attacks Based on Hardware Characteristics, Patent No. 2022113990841, CN115862670B, Granted: 2026-01-09, Rank 1/8
- Method for Designing Voice Jamming Noise Based on Human Speech Structure, Patent No. 2022114278110, Granted: 2026-03-02, Rank 1/8
- Camera Fingerprint Privacy Protection Method against Linkage and Forgery Attacks, Patent No. 202211674524X, Granted: 2025-11-28, Rank 1/8
- Security Evaluation Method for Speech Recognition Models via Semantic Space Perturbation, Patent No. 2023110087782, CN116758899B, Granted: 2023-10-13, Rank 1/7
- Enhanced Deepfake Image Detection Method and Device Based on Generative Adversarial Networks (GANs), Patent No. 2024100814592, CN117593311B, Granted: 2024-06-21
- Cross-Domain Detection Method and Device for Deep Synthetic Audio Based on Self-Supervised Auxiliary Tasks, Patent No. 2025100324916, CN 119479611 B, Granted: 2025-04-29, Rank 3/5
- Privacy Protection Method, System, and Medium for Voice Communication Based on Voice Obfuscation, Patent No. 2025100324174, CN 119449493 B, Granted: 2025-07-08, Rank 2/5
- Review System, Method, Computer Device, and Medium for Image Sensitive Elements, Patent No. 2025101488133, CN119625437B, Granted: 2025-04-22, Rank 4/7
- AIGC Image Watermarking Method and System via Diffusion-Model Generation-Path Deflection, Patent No. 2025108297832, CN 120339030 B, Granted: 2025-08-26, Rank 3/7
- Privacy-Protecting Noise Generation Method Based on Black-Box Adversarial Examples, Patent No. 2025112567923, CN120748435B, Granted: 2025-11-04, Rank 3/5
- Watermark Anti-Counterfeiting Evaluation Method, System, Electronic Device, and Storage Medium Based on Diffusion Models, Patent No. 202511477887.8, CN120931468B, Granted: 2026-01, Rank 3/5
- Cross-Domain Adaptive Deepfake Speech Detection Method, Patent No. 2022115339855, CN116153331A, Granted: 2023-05-23, Rank 1/8

Other Academic Activities

- Contributed to the preparation of a €720,839 grant proposal (Science Foundation Ireland — SFI) on Security and Privacy of Personal Voice Assistants under Prof. Utz Roedig (2019)
- Visiting Scholar at the Department of Computer Science, University College Cork (UCC), Ireland (2019—2020)
- Participated in the São Paulo Advanced Science School (ESPCA) on Smart Cities, hosted by the University of São Paulo (2017); selected as one of 75 global top graduate students and postdoctoral researchers (sponsored by FAPESP)

Impacts

Industry Contributions

- Proposed SurrogatePrompt, a prompt attack method to bypass safeguards of commercial large-scale AI models (e.g., Midjourney, Stability.ai), enabling systematic generation of policy-violating content. Findings were acknowledged by these leading vendors.

- Developed WMCopier, an adversarial watermark forgery method that compromises commercial watermarking systems; recognized by Amazon's Responsible AI Team for identifying critical vulnerabilities (official thank-you letter).
- Developed and open-sourced ALIF, the first black-box adversarial attack framework leveraging linguistic features; adopted by NVIDIA for integration into their official AI security toolkit.

Media Recognition

- SonarSnoop: Active Acoustic Side-Channel Attacks — garnered attention from IT media (Motherboard, ZDNet, Sophos); praised by Bruce Schneier and Prof. Ross Anderson.

Google Scholar

`https://scholar.google.com/citations?user=hz3kfCEAAAAJ&hl=en`